

GitHub repository where all documentation & code can be found. [billbill100 \(github.com\)](https://github.com/billbill100)

ESP32 Shutter Tester pcb V1.1 06/08/2026

As an alternative to using the expansion board to mount the ESP32 module, a bespoke pcb has been designed.

It is very similar in design to the expansion board, but has the advantage that both the ESP32 module and the tft screen plug directly into the pcb, with all required connections for the tft routed directly on the pcb.

It also has additional 3, 5 and 0v connectors, making wiring simpler.

All of the other connections for sensors etc are broken out onto screw-terminals in the same way as the expansion board.

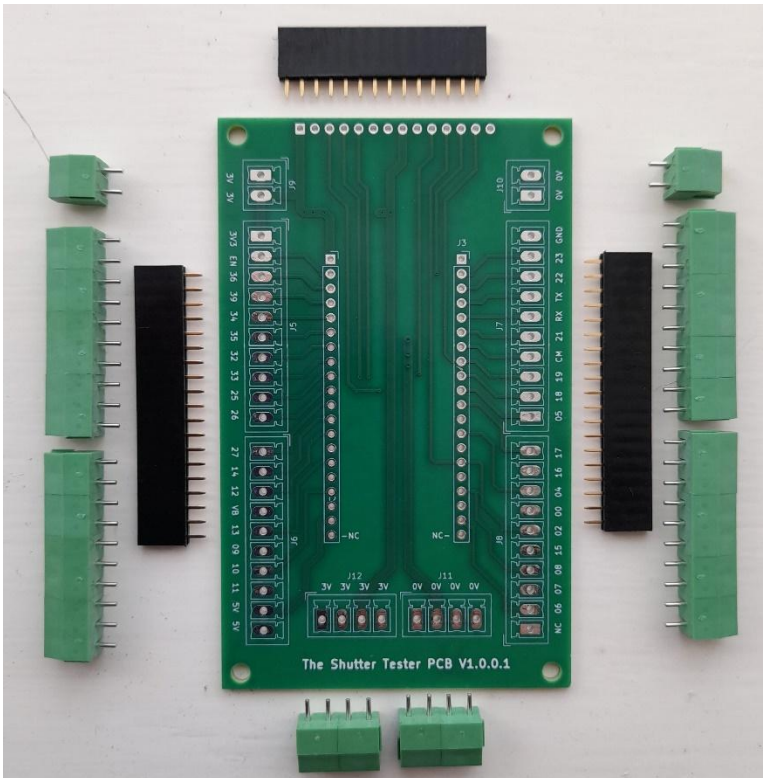
Other than this, wiring and using the pcb is identical to that of the expansion board.

To obtain a pcb, it can be ordered from pcbway, who offer a good cheap pcb fabrication service.

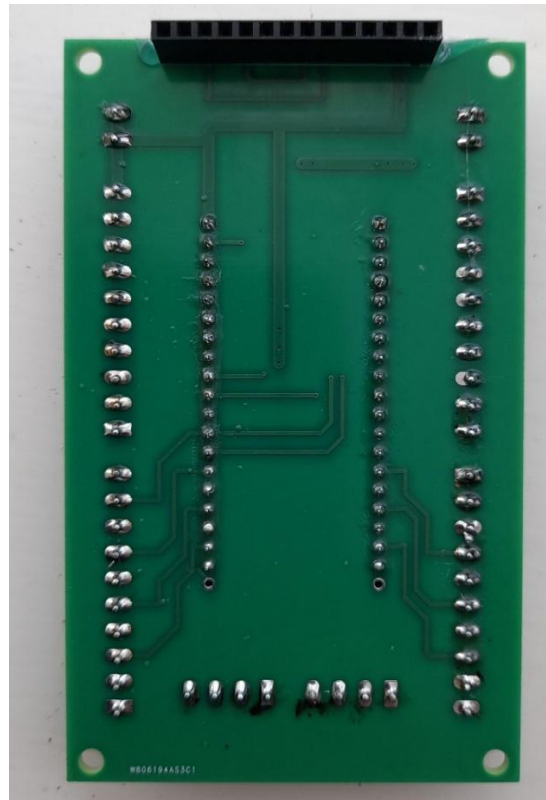
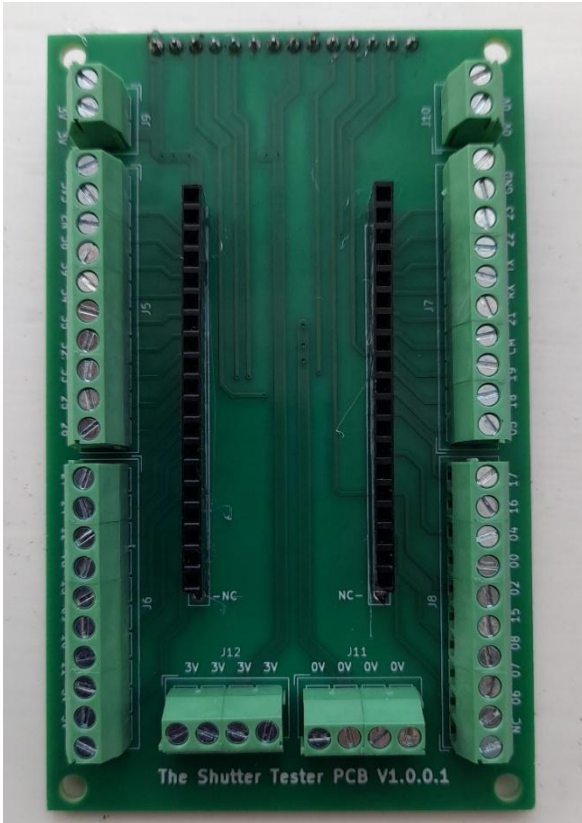
They do however have a minimum order quantity (5 or 10) so it is well worth asking on Photrio forum if anybody has any spare pcb to sell. They may also have spare header sockets and screw terminal.

The pcb requires header sockets and screw terminals and must be assembled and soldered by the user.

Note- Pictures show the V1 board. It is designed to fit the narrower ESP32 dev board specified in the parts document. A newer board now available also allows the wider ESP32 dev board to fit.



pcb and additional parts required parts.



Completed pcb, showing each side

Extra 3v and 0v screw terminals can be seen at the bottom centre of the pcb

Extra 3v and 0v screw terminals top left and right.

Extra 5v screw terminal bottom left.

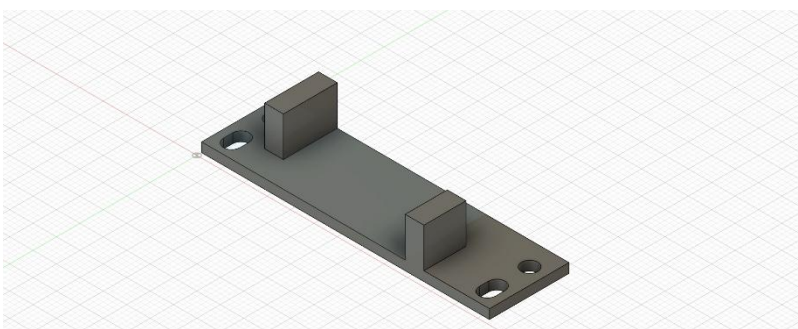
Important Note:- Ensure the tft connector is soldered on the rear side of the pcb
and the ESP32 and screw terminals are soldered on the front side of the pcb.

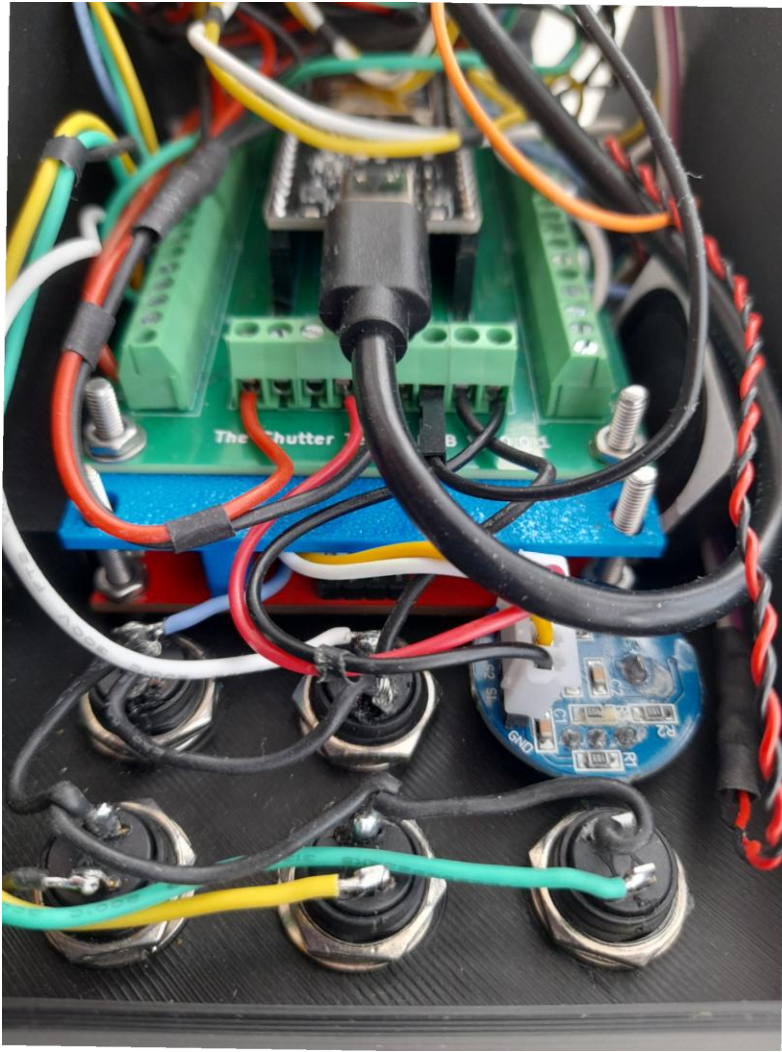
The pre-drilled holes at the top of the pcb align with the holes in the tft, and can be secured using the same tft mounting bolts.

M3 bolts & nuts can be used in the bottom holes, as stand-offs. With the heads hot-glued to the tft pcb.
This method is shown in the 'Build a Box' document

A better solution is to 3d print the tft pcb spacer.

This bolts onto the underside of the pcb, with the rectangular stand-offs facing down, (using the round holes on the 3d printed part) using M3 X12mm bolts or similar.
with the elongated holes fitting onto the tft mounting bolts.





Pcb spacer, printed in blue, fitted in position.

Parts required for the pcb

Screw terminals, 3.5mm spacing.

[100PCS 3.5mm 2P 3P Screw Terminal Block Connector KF350](#)

Screw terminals, 3.5mm spacing.

They come in 2 or 3 way and slide together to form larger ways.

4 x 10-way, 2 x 4-way and 2 x 2-way are required.

So one pack of 2-way connectors are required, which is enough for three boards.

(buy 1 pack of 2-way and one pack of 3-way, which will be plenty for 10 boards)



Header sockets, female.

[10PCS 2.54mm Single Row Female Header 1x2p to 1x40p Pin Connector](#)

2 of 19-way required and
1 of 14-way required.

They come in packs of 10, so one pack of the 19-way will make 5 boards.
(buy 2 packs of 19-way to make 10 boards)

